

Boundary-Inversion Cosmology VI: A Sqrt-Reg EFT-Clock Inflationary Branch and Calibration-Frame Late-Universe Extension

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Abstract

We present Boundary-Inversion Cosmology VI (BIH-VI) as a boundary-anchored effective-field-theory cosmology built around the invariant marginal-horizon seam $\chi = h^{ab}\nabla_a R \nabla_b R = 0$, equivalently $R = 2GE_{\text{MS}}$. The early-universe branch realizes a Sqrt-Reg entropy-selected density shield through a stable EFT clock with background evolution $H^2 = H_c^2 y(2-y)$, $y_N = -m(y)y$, and $\epsilon_H = m(1-y)/(2-y)$. For the frozen benchmark $m(y) = \mu(1-y)^3$, $\mu = 1$, $c_s = 1$, and $N_* = 55$, the model gives $n_s \simeq 0.9652$, $r \simeq 1.4 \times 10^{-3}$, and $\alpha_s \simeq -5.8 \times 10^{-4}$, with stable scalar and tensor quadratic actions. The exit switch is pivot-safe and produces a localized ultra-small-scale feature mapped to $k \sim 1.3 \times 10^{23} \text{ Mpc}^{-1}$, or $f \sim 200 \text{ MHz}$, while remaining quiet in ordinary CMB/LSS channels.

We then test whether the same boundary-centered framework can be extended to late-universe cosmology without modifying the frozen inflationary branch. The low-density expansion of the Sqrt-Reg background shows that the entropy shield acts as a UV density regulator rather than an infrared dark-energy mechanism. A direct metric-lapse interpretation, $H_{\text{obs}}(z) = \nu(z)H_{\text{geom}}(z)$, is rejected at the diagnostic level because the required local uplift produces unacceptable supernova-shape and BAO-distance distortions. We therefore define a calibration-frame late branch in which $\nu_{\text{metric}}(z) = 1$ for light-propagation distance integrals, while the local distance ladder may measure a substrate-dependent calibration offset $H_0^{\text{ladder}} = \nu_{\text{cal}}H_0^{\text{geom}}$, equivalently $\Delta M_B = 5 \log_{10} \nu_{\text{cal}} \simeq 0.18 \text{ mag}$ for the current SH0ES/Planck-scale target.

The late branch is formulated as a falsifiable likelihood architecture rather than a completed solution. Posterior samples are translated into a substrate-exchange diagnostic Q_{bulk} through $\dot{\rho}_{\text{DE}} + 3H(1 + w_{\text{int}})\rho_{\text{DE}} = Q_{\text{bulk}}$, with $Q_{\text{bulk}}/(H\rho_{\text{DE}}) = 3(w_{\text{int}} - w_{\text{eff}})$. The C3 branch survives only if the reconstructed exchange history is smooth, bounded, and low-frequency. BIH-VI is therefore presented as a stable EFT-level early-universe branch plus a calibration-frame late-universe validation program, not as a UV-complete microscopic substrate theory.

Scope boundary. BIH-VI is presented as an EFT-level cosmological framework. It does not claim to be a UV-complete microscopic theory. The early-universe branch is frozen at the Sqrt-Reg/EFT-clock benchmark. The late-universe branch is a diagnostic calibration-frame extension motivated by the failure of universal metric-lapse deformations. The calibration-frame branch is falsifiable through full distance-likelihood tests and through the smoothness, boundedness, and physical interpretability of the reconstructed substrate-exchange history $Q_{\text{bulk}}(z)$.

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1 Introduction

The standard cosmological model has achieved remarkable empirical success, but two observational fronts motivate controlled extensions. The first is the early-universe problem: a viable inflationary or pre-inflationary framework must produce a nearly scale-invariant scalar spectrum, satisfy tensor constraints, avoid obvious perturbative instabilities, and remain falsifiable. The second is the late-universe calibration problem: the expansion history inferred from early-universe data remains in tension with local distance-ladder determinations of H_0 , and any proposed mechanism must survive the joint pressure of supernova luminosity distances, BAO geometry, and the CMB acoustic scale.

BIH-VI addresses these questions as an effective-field-theory framework organized around causal boundaries. Its geometric core is the spherical marginal-horizon condition

$$\chi = h^{ab}\nabla_a R \nabla_b R = 0, \quad (1)$$

equivalently

$$R = 2GE_{\text{MS}}, \quad (2)$$

where R is the areal radius and E_{MS} is the Misner-Sharp energy. This condition unifies black-hole trapped and cosmological anti-trapped sign sectors at the level of spherical-horizon geometry. The distinction between collapse and expansion is not encoded in χ alone, but in the orientation of the future null expansions.

Paper hierarchy. The paper is organized as follows. Sections 2–4 summarize the frozen early branch. Section 5 documents the late-universe stress test and the rejection of the universal metric-lapse branch. Section 6 defines the calibration-frame extension. Section 8 gives the substrate-exchange falsifiability test. Appendices collect equations, rejected branches, likelihood scaffolding, repository structure, and open work.

2 Boundary seam and Sqrt-Reg thermodynamics

For a spherically symmetric spacetime,

$$ds^2 = h_{ab}dx^a dx^b + R^2 d\Omega^2, \quad (3)$$

the future-directed radial null expansions are

$$\theta_{\pm} = \frac{2}{R}\ell_{\pm}^a \nabla_a R. \quad (4)$$

With cross-normalization $\ell_+ \cdot \ell_- = -2$, one has

$$\theta_+ \theta_- = -\frac{4\chi}{R^2}. \quad (5)$$

Thus $\chi > 0$ corresponds to normal sign sectors, $\chi = 0$ to marginal surfaces, and $\chi < 0$ to trapped or anti-trapped regions. The sign pair distinguishes black-hole trapped $(-, -)$ from cosmological anti-trapped $(+, +)$ regions.

Apparent-horizon thermodynamics is applied on the cosmological side using

$$x = H^2 + \frac{k}{a^2}, \quad R_A = x^{-1/2}, \quad A = \frac{4\pi}{x}. \quad (6)$$

The Sqrt-Reg entropy functional is

$$S_{\sqrt{}}(A) = \frac{1}{4G} \left[\sqrt{A(A - A_c)} + A_c \operatorname{arccosh} \sqrt{\frac{A}{A_c}} \right], \quad (7)$$

with

$$S_A = \frac{1}{4G\sqrt{1 - A_c/A}}, \quad G_{\text{eff}} = \frac{1}{4S_A} = G\sqrt{1 - \frac{A_c}{A}}. \quad (8)$$

The associated Friedmann function is

$$F_{\sqrt{}}(x) = 2x_c \left[1 - \sqrt{1 - x/x_c} \right], \quad (9)$$

so that in the flat case

$$F_{\sqrt{}}(H^2) = \frac{8\pi G}{3} \rho. \quad (10)$$

This gives a density shield

$$H^2 = \frac{8\pi G}{3} \rho \left[1 - \frac{2\pi G \rho}{3H_c^2} \right], \quad \rho_{\text{max}} = \frac{3H_c^2}{4\pi G}. \quad (11)$$

3 Frozen EFT-clock branch

The stable branch realizes the Sqrt-Reg background inside the EFT of cosmological perturbations:

$$S_{\text{EFT}} = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \Lambda(t) - c(t)g^{00} + \frac{M_2^4(t)}{2} (\delta g^{00})^2 + \dots \right]. \quad (12)$$

The background is

$$H^2 = H_c^2 y(2 - y), \quad y_N = -m(y)y, \quad (13)$$

with

$$\dot{H} = -mH_c^2 y(1 - y), \quad \epsilon_H = \frac{m(1 - y)}{2 - y}. \quad (14)$$

The EFT functions are

$$c(y) = M_{\text{Pl}}^2 m H_c^2 y(1 - y), \quad (15)$$

$$\Lambda(y) = M_{\text{Pl}}^2 H_c^2 y [(6 - m) + (m - 3)y], \quad (16)$$

$$M_2^4(y) = \frac{c(y)}{2} (c_s^{-2} - 1). \quad (17)$$

The quadratic scalar and tensor actions are

$$S_{\zeta}^{(2)} = M_{\text{Pl}}^2 \int dt d^3x a^3 \frac{\epsilon_H}{c_s^2} \left[\dot{\zeta}^2 - c_s^2 \frac{(\nabla \zeta)^2}{a^2} \right], \quad (18)$$

$$S_T^{(2)} = \frac{M_{\text{Pl}}^2}{8} \int dt d^3x a^3 \left[\dot{h}_{ij}^2 - \frac{(\nabla h_{ij})^2}{a^2} \right]. \quad (19)$$

The tested stability conditions are $\epsilon_H > 0$, $c_s^2 > 0$, and $c_T^2 = 1$.

3.1 Cubic gear and benchmark observables

Define $z = 1 - y$ and

$$m(y) = \mu z^p. \quad (20)$$

The benchmark used throughout the frozen branch is

$$p = 3, \quad \mu = 1, \quad c_s = 1, \quad N_* = 55, \quad z_e = 1/3. \quad (21)$$

The benchmark gives

$$n_s \simeq 0.9652, \quad r \simeq 1.4 \times 10^{-3}, \quad \alpha_s \simeq -5.8 \times 10^{-4}. \quad (22)$$

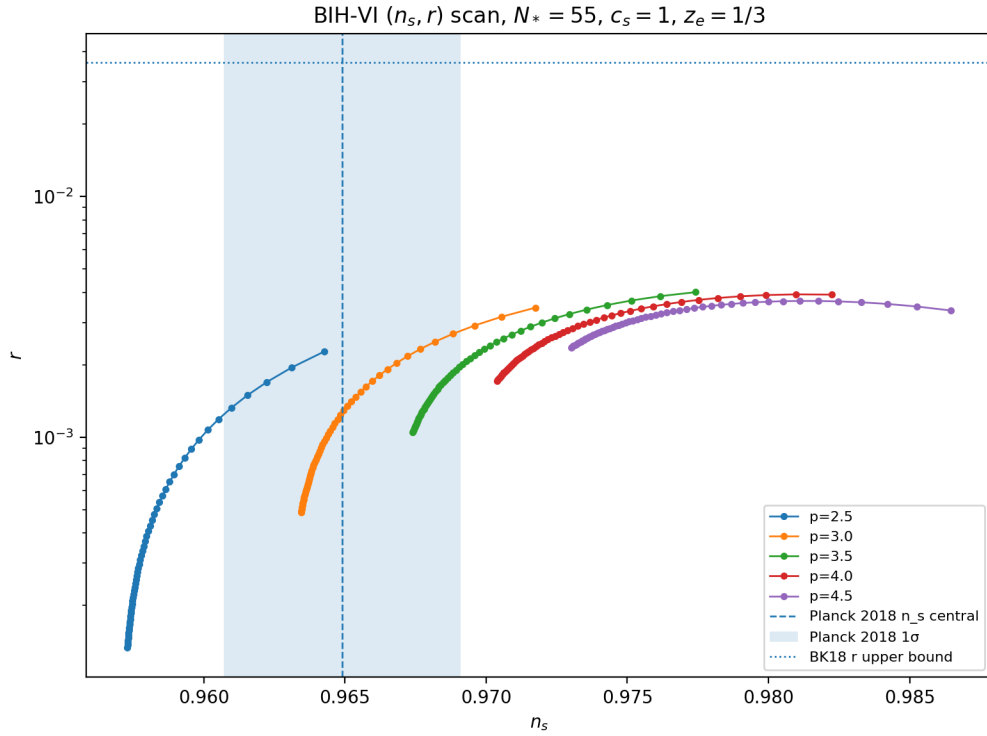


Figure 1: Frozen early-branch parameter island in the (n_s, r) plane. Replace with final publication figure and caption.

4 Exit switch and ultra-small-scale relic

The switch is

$$m(y) = \mu(1 - y)^3 + m_{\text{post}}S(y), \quad S(y) = \frac{1}{1 + \exp[(y - y_g)/\Delta]}. \quad (23)$$

For the benchmark $y_g = 2/3$, $m_{\text{post}} = 4$, and $\Delta = 0.01$, the switch remains pivot-safe while producing a localized exit feature.

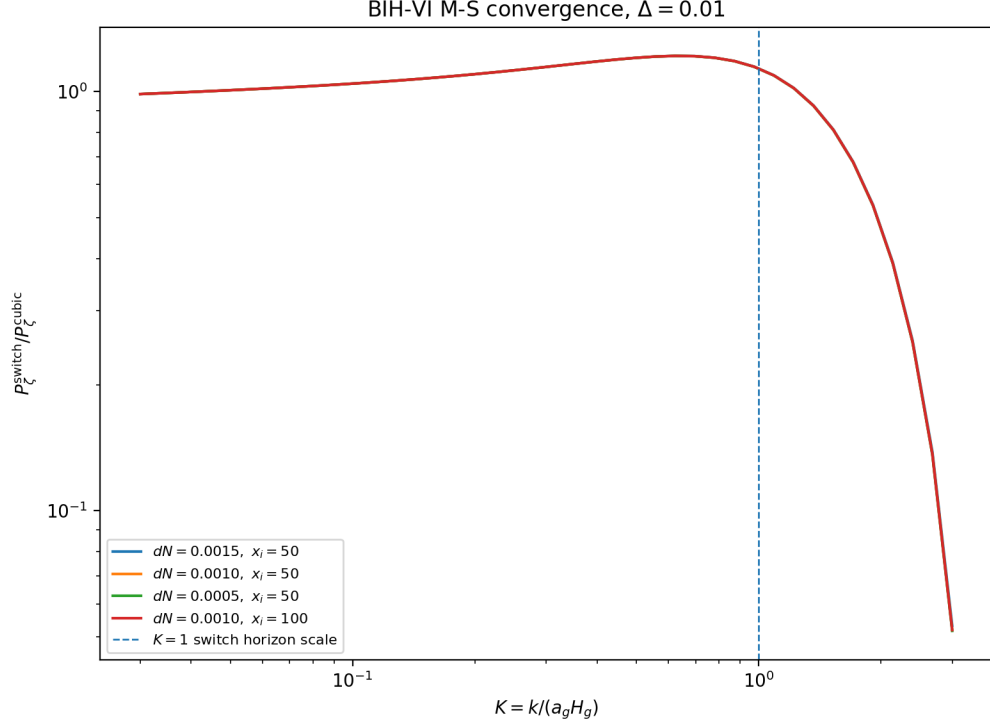


Figure 2: Mukhanov-Sasaki convergence for the exit switch at $\Delta = 0.01$.

The leading feature bispectrum diagnostic uses the switch-enhanced interaction

$$S_3^{\text{switch}} \supset \frac{M_{\text{Pl}}^2}{2} \int dt d^3x a^3 \epsilon_H \dot{\eta}_H \zeta^2 \dot{\zeta}. \quad (24)$$

The normalized feature amplitude can reach order 10^3 near the exit feature, but only where the scalar power is deeply suppressed; the absolute bispectrum proxy remains finite.

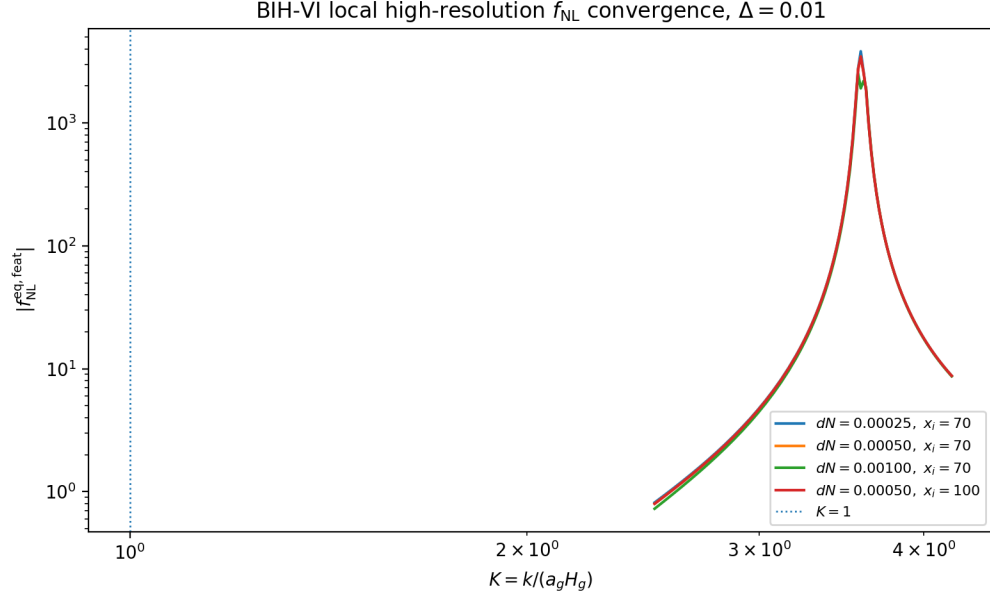


Figure 3: High-resolution equilateral feature-bispectrum diagnostic near the exit feature.

The scale map places the feature at

$$k_{\text{peak}} \sim 1.3 \times 10^{23} \text{ Mpc}^{-1}, \quad f_{\text{peak}} \sim 2.0 \times 10^8 \text{ Hz}. \quad (25)$$

Thus the exit feature is not a conventional CMB, LSS, or ordinary 21-cm target.

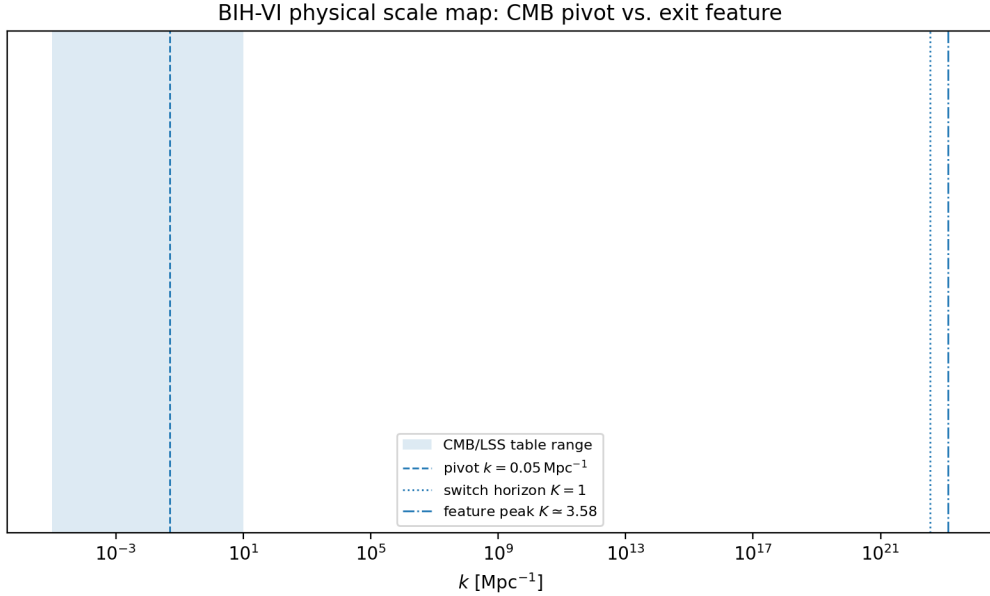


Figure 4: Physical scale map for the exit feature.

5 Late-universe diagnostic gate

Having established the stable EFT-clock realization of the Sqrt-Reg density shield as an early-universe branch, we now assess whether the same boundary-centered framework can be extended to late-time cosmology without modifying its frozen inflationary mechanics. The low-density expansion of the Sqrt-Reg background shows that the entropy shield acts as a UV regulator rather than an infrared dark-energy mechanism. We therefore treat the late universe as a separate observational gate on the BIH clock/stress sector. A direct metric-lapse interpretation, in which a redshift-dependent clock factor multiplies all distance integrals, is rejected at the diagnostic level by supernova-shape and BAO-geometry constraints. Consequently, we define a calibration-frame branch: the global geometry remains governed by $H_{\text{geom}}(z)$, while the local distance ladder is allowed to measure a substrate-dependent calibration offset $\Delta M_B \simeq 0.18 \text{ mag}$. This recasts the Hubble tension as a testable calibration-frame hypothesis rather than a universal modification of light-propagation distances.

5.1 Low-density Sqrt-Reg limit

For $A \gg A_c$, equivalently $x \ll x_c$,

$$S_A = \frac{1}{4G} \left[1 + \frac{1}{2} \frac{x}{x_c} + \frac{3}{8} \left(\frac{x}{x_c} \right)^2 + \dots \right], \quad (26)$$

so the correction decays in the infrared. The Sqrt-Reg shield is therefore a high-density regulator, not a standalone late dark-energy mechanism.

5.2 Metric-lapse branch and rejection

The rejected late branch is

$$H_{\text{obs}}(z) = \nu(z) H_{\text{geom}}(z). \quad (27)$$

If $\nu(0)$ is forced toward the local distance-ladder uplift, the same lapse enters the distance integrals

$$D_C(z) = c \int_0^z \frac{dz'}{\nu(z') H_{\text{geom}}(z')}, \quad (28)$$

and creates supernova and BAO strain.

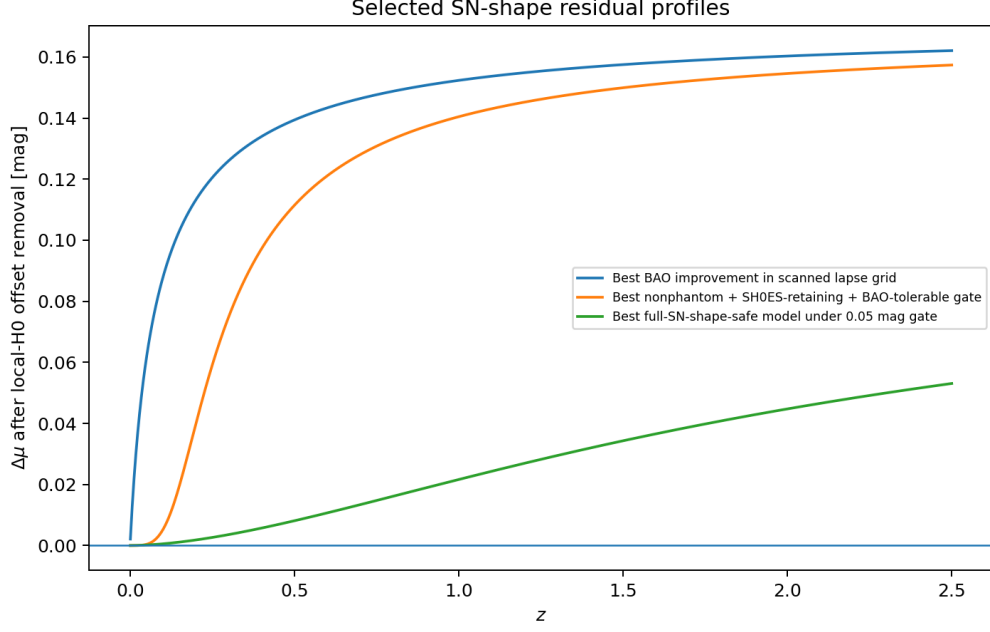


Figure 5: Diagnostic SN-shape residuals from the metric-lapse branch. This branch should be described as rejected at the diagnostic level unless a future formulation changes the frame interpretation.

6 Calibration-frame extension

The surviving C3 branch imposes

$$\nu_{\text{metric}}(z) = 1 \quad (29)$$

inside BAO, SN, and CMB distance integrals. The local ladder instead measures

$$H_0^{\text{ladder}} = \nu_{\text{cal}} H_0^{\text{geom}}. \quad (30)$$

Equivalently,

$$\Delta M_B = 5 \log_{10} \nu_{\text{cal}}. \quad (31)$$

For $H_0^{\text{ladder}} = 73.17 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $H_0^{\text{geom}} = 67.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$,

$$\nu_{\text{cal}} \simeq 1.08625, \quad \Delta M_B \simeq 0.1797 \text{ mag}. \quad (32)$$

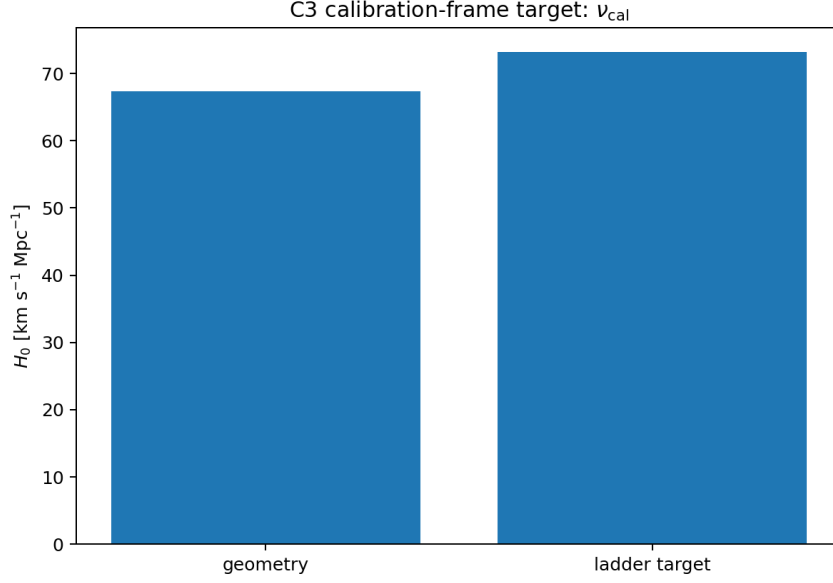


Figure 6: Calibration-frame target: the local ladder measures a readout offset rather than a universal metric-distance deformation.

6.1 Geometry scaffold

The geometry sector may be scaffolded with CPL:

$$w_{\text{eff}}(z) = w_0 + w_a \frac{z}{1+z}, \quad (33)$$

$$E_{\text{BIH}}^2(z) = \Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_k(1+z)^2 + \Omega_{\text{DE},0} f_{\text{DE}}(z), \quad (34)$$

with

$$f_{\text{DE}}(z) = (1+z)^{3(1+w_0+w_a)} \exp \left[-3w_a \frac{z}{1+z} \right]. \quad (35)$$

The distances use $H_{\text{geom}}(z) = H_0^{\text{geom}} E_{\text{BIH}}(z)$.

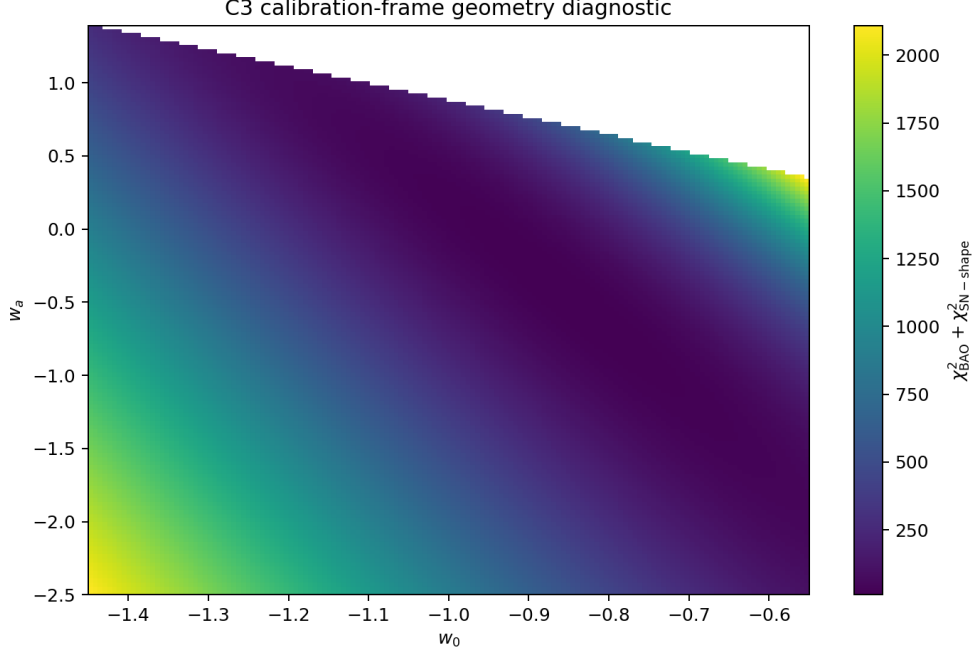


Figure 7: C3 calibration-frame geometry diagnostic in the CPL scaffold. Production likelihood results should replace this diagnostic plot.

7 Likelihood architecture

The C3 production parameter vector is

$$\Theta_{C3} = \{h, \Omega_m, \Omega_k, \omega_b, w_0, w_a, M_B^{\text{geom}}, \Delta M_B\}. \quad (36)$$

The likelihood must include DESI DR2 BAO, full Pantheon+SH0ES covariance, a compressed or full CMB acoustic-scale prior, and independent calibration/anchor priors. This section should be updated once the production chain has been run.

Sector	Parameter(s)	Role
Geometry	$h, \Omega_m, \Omega_k, \omega_b, w_0, w_a$	BAO/SN/CMB distances
Calibration	$M_B^{\text{geom}}, \Delta M_B$	local readout offset
Falsifier	$Q_{\text{bulk}}(z)$	thermodynamic consistency

Table 1: C3 production-likelihood parameter hierarchy.

8 Substrate-exchange reconstruction

The diagnostic dark-sector exchange equation is

$$\dot{\rho}_{\text{DE}} + 3H(1 + w_{\text{int}})\rho_{\text{DE}} = Q_{\text{bulk}}. \quad (37)$$

If the fitted background is represented by $w_{\text{eff}}(z)$, then

$$\frac{Q_{\text{bulk}}}{H\rho_{\text{DE}}} = 3(w_{\text{int}} - w_{\text{eff}}). \quad (38)$$

For $w_{\text{int}} = -1$,

$$\frac{Q_{\text{bulk}}}{H\rho_{\text{DE}}} = -3[1 + w_{\text{eff}}(z)]. \quad (39)$$

The C3 branch is viable only if the posterior-implied exchange profile is smooth, bounded, low-frequency, and physically interpretable.

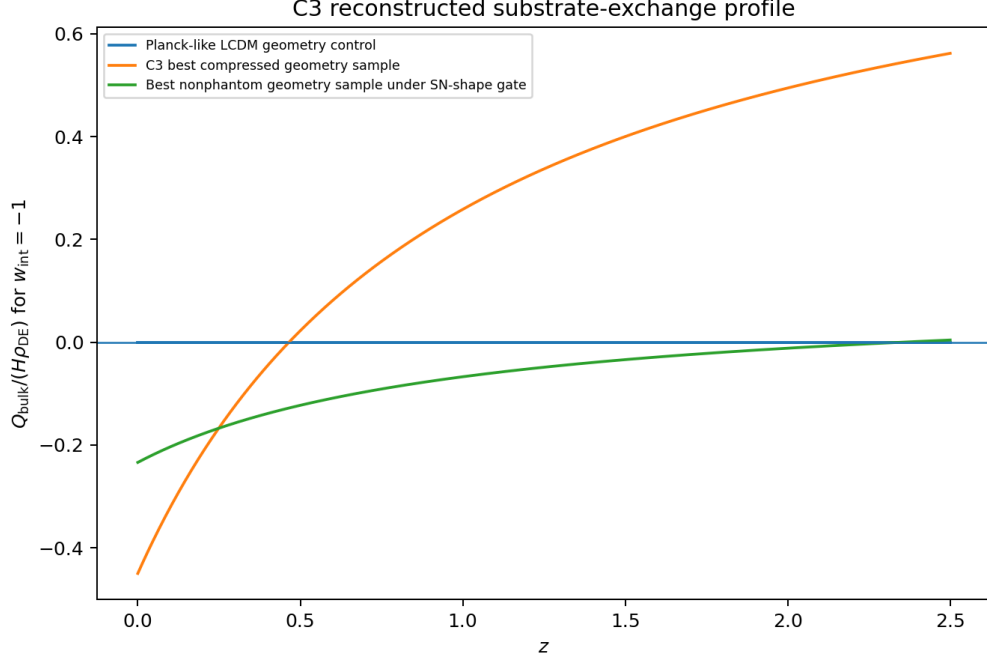


Figure 8: Selected $Q_{\text{bulk}}/(H\rho_{\text{DE}})$ profiles from the C3 diagnostic scaffold.

9 Falsifiability and future tests

The framework is falsifiable through at least six routes:

1. Future tensor limits push well below $r \sim 10^{-3}$ without detection and without a viable parameter island.
2. The preferred scalar tilt moves outside the BIH-VI early-branch island after nuisance and reheating freedom are included.
3. Running is observed with a large or opposite-sign value relative to the benchmark.
4. Small-scale probes rule out the predicted localized suppression/ringing region.
5. Full C3 likelihoods reject $\Delta M_B \simeq 0.18$ mag once independent anchors are included.
6. The reconstructed $Q_{\text{bulk}}(z)$ is jagged, discontinuous, or requires high-frequency sign-flip chatter.

10 Conclusion

We have presented BIH-VI as a boundary-anchored EFT-clock cosmological framework organized around the invariant marginal-horizon seam $\chi = 0$, equivalently $R = 2GE_{\text{MS}}$. The frozen early-universe branch realizes a Sqrt-Reg density shield with stable scalar and tensor quadratic actions and a Planck-compatible observable island. The exit switch is pivot-safe and maps to an ultra-small-scale, UHF-relic-quiet feature.

The late-universe result is deliberately hierarchical. A universal metric-lapse interpretation fails diagnostic distance-gate tests and is rejected as the primary H_0 mechanism. The surviving C3 branch treats the Hubble discrepancy as a calibration-frame hypothesis: metric distances remain governed by $H_{\text{geom}}(z)$, while the local distance ladder may measure a calibration/readout offset $\Delta M_B \simeq 0.18$ mag. This is not yet a derived solution of the Hubble tension. It is a falsifiable late-universe extension whose viability depends on production likelihoods, independent anchors, and the smoothness of reconstructed $Q_{\text{bulk}}(z)$.

A Equation map

A.1 Geometric seam

$$\chi = h^{ab} \nabla_a R \nabla_b R, \quad (40)$$

$$\chi = 1 - \frac{2GE_{\text{MS}}}{R}, \quad (41)$$

$$\chi = 0 \iff R = 2GE_{\text{MS}}. \quad (42)$$

A.2 Early branch

$$H^2 = H_c^2 y(2 - y), \quad y_N = -m(y)y, \quad \epsilon_H = \frac{m(1 - y)}{2 - y}. \quad (43)$$

A.3 Late C3 branch

$$\nu_{\text{metric}}(z) = 1, \quad H_0^{\text{ladder}} = \nu_{\text{cal}} H_0^{\text{geom}}, \quad \Delta M_B = 5 \log_{10} \nu_{\text{cal}}. \quad (44)$$

B Rejected branches and negative results

Document here: log-critical entropy branch, pure $f(T)$ action, minimal $f(T, B)$ repair, analytic $f(R)$ branch, C1 metric-lapse branch, and C2 metric-lapse+CPL branch. This appendix should include failure criteria rather than rhetorical summaries.

C Likelihood implementation

Insert repository link and code usage once public. Include Cobaya wrapper, data-preparation instructions, Pantheon+SH0ES covariance handling, DESI DR2 BAO vector/covariance, CMB acoustic-scale prior, and convergence criteria.

D Production tables to insert

1. Early benchmark table: $(\mu, z_*, y_*, n_s, r, \alpha_s, H_c/M_{\text{Pl}})$.
2. C1/C2 distance-gate table.
3. C3 posterior constraints.
4. Q_{bulk} smoothness metrics.
5. Independent-anchor sensitivity table.

E Repository structure

Recommended supplemental repository layout:

```
BIH-VI/  
  paper/  
  theory/  
  likelihoods/cobaya/  
  likelihoods/python/  
  data/  
  notebooks/  
  results/  
  figures/  
  archive/rejected_branches/
```

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